

Single-Pass Drone Video to 3D on One GPU: Overlap-Band Keyframes, a Parallax Test for “Is It 3D?”, Monocular Depth Priors and Gaussian Splatting

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Abstract

We present an end-to-end system that turns a single drone pass — one video, no flight planning, no ground control — into a 3D Gaussian Splatting model, a textured mesh and, when a flight log is available, geo-referenced metric outputs. Every stage runs on one GPU: hardware video decoding, a keyframe selector built on dense optical flow, GPU structure from motion, a diffusion-transformer depth prior and splat training. The selector formalises the intuition that consecutive views must be *similar enough to match and different enough to triangulate*: it chains optical flow to measure co-visibility and picks the sharpest frame inside an overlap band $[\tau, \tau + \delta]$, so every surface point is seen by about $1/(1 - \tau)$ keyframes. Before any reconstruction it answers whether a pass contains recoverable 3D structure at all, using rotation-compensated parallax measured with *direct* keyframe-pair flow and normalised by an estimated noise floor; chained tracks are unsuitable because their error grows linearly with the gap and masquerades as parallax. On edited 4K footage the selector finds the individual camera moves, and the keyframes it chooses all register. On a straight single-pass track we show that the usual 7-DoF GPS alignment silently rolls the model about the flight axis; levelling on the ground plane and fitting four degrees of freedom removes the failure and keeps metric scale error below 0.6% in simulation. All numbers are produced by scripts in the repository from the run directories.

1 Introduction

Operational mapping — disaster response, inspection, reconnaissance — often has one chance to fly over a site. Classical photogrammetry assumes the opposite: planned grids, high overlap, several passes and ground control points. Problem statement 26158 asks for an accurate, textured, georeferenced 3D model from a *single pass* of a moving UAV [1].

A single pass is hard for three distinct reasons, and each one maps to a component of our system.

1. **Which frames?** A video carries thousands of highly redundant frames and a few bad ones (motion blur, compression, fades). Sampling at a fixed rate is the common default; it either wastes compute or, where the drone moves fast, leaves gaps that break the reconstruction into pieces (§3.3).

2. **Is it 3D at all?** A drone that yaws on the spot, or looks at a distant or planar scene, produces video in which a homography explains every frame pair. No amount of computation recovers depth from such footage; the system should say so before spending it (§3.4).
3. **Where is it?** A single pass is nearly a straight line. Aligning the model to GPS through camera centres alone leaves the roll about that line undetermined (§3.8).

Around these we assemble a GPU pipeline: NVDEC decoding, RAFT optical flow [2], spirula-studio’s GPU structure from motion and 3D Gaussian Splatting [3, 4], and the Marigold v2 depth prior [5] aligned to, and scored against, the SfM points.

Contributions. (i) An overlap-band keyframe selector over chained dense flow with an explicit multiplicity model, and pass segmentation for edited footage. (ii) A per-pass 3D verdict from direct-flow residual parallax with a data-driven noise floor, validated on a synthetic pure-rotation control; we report where Torr’s GRIC [6] fails on flow correspondences. (iii) A measurement of chained versus direct optical-flow correspondence error against exact ground truth. (iv) A ground-levelled 4-DoF georeferencing rule with a significance test for sloped terrain, leave-one-out error and a jackknife scale uncertainty. (v) An engineering study of keeping an A100 busy: removing host synchronisation from RAFT, CUDA-Graph replay, GPU colour conversion and nvJPEG/NVENC, with utilisation recorded for every stage.

2 Related work

Structure from motion and splatting. Incremental SfM as in COLMAP [7] remains the reference for camera recovery; spirula-studio re-implements an incremental mapper on the GPU and adds sequence-aware matching for video [3]. 3D Gaussian Splatting [4] represents a scene as anisotropic Gaussians optimised by differentiable rasterisation; gsplat [8] provides the rasteriser we use for fly-through rendering.

Keyframe selection. Visual SLAM systems insert keyframes by co-visibility and tracking quality,

and ORB-SLAM chooses between homography and fundamental-matrix initialisation by model scores [9]; Torr’s GRIC formalises that choice [6]. We use dense flow with the forward–backward consistency test of Sundaram et al. [10].

Monocular depth. Marigold [11] repurposed a latent diffusion model for affine-invariant depth; Marigold v2 [5] fine-tunes the Qwen-Image-Edit diffusion transformer [12] to predict affine-invariant log depth in one step. Depth Anything V2 [13] is a strong discriminative alternative.

Compositional reconstruction. WorldSculpt [14] lifts per-object priors into cluttered scenes from posed, masked views; we discuss it as a downstream extension in §5.

3 Method

3.1 GPU ingest

Frames are decoded by NVDEC through `ffmpeg` (`-hwaccel cuda`), resized on the GPU (`scale_cuda`) and leave the decoder as raw NV12; conversion to RGB (BT.709, limited range) runs in PyTorch on the GPU. A background thread streams chunks so decoding overlaps optical flow. Keyframes are later re-decoded at full resolution, selected by timestamp in groups of 24 on two concurrent decoder sessions, cropped on the GPU and encoded with nvJPEG.

3.2 Pass segmentation

For analysis frames $I_0, \dots, I_{N-1}$ (long side 640 px, ≈ 12 fps) we compute RAFT flow $\mathbf{f}_t$ from t to $t+1$ and $\mathbf{b}_t$ back, and the share c_t of pixels passing the forward–backward test $\|\mathbf{f} + \mathbf{b}\|^2 < 0.01(\|\mathbf{f}\|^2 + \|\mathbf{b}\|^2) + 0.5$ [10]. Inside a continuous camera move c_t stays high; a cut, fade or dissolve collapses it. A *pass* is a maximal run with $c_t \geq 0.35$, no dark or flat frame, and at least 2 s; its ends are trimmed while they are darker than $0.8\times$ the pass median, removing fade-ins and fade-outs (a fade kills tracks, so without trimming it attracts keyframes, and a fade frame is a meaningless held-out view). Letter-box bars are found before analysis from the per-row and per-column maximum luma over twelve seek-sampled frames and cropped on the GPU: static black would otherwise count as perfectly tracked content, inflating overlap, and as trivially correct pixels, inflating held-out PSNR. Real footage is often edited; a true single pass yields one pass, and SfM rejoins passes that turn out to share views.

3.3 Overlap-band keyframes

A grid G of points (8 px spacing) on keyframe K is chained through the consecutive flows; a track dies when it fails the consistency test or leaves the frame. With $A_t \subseteq G$ the tracks alive at frame t ,

$$\text{ov}(K, t) = \min \left(\frac{|A_t|}{|G|}, \frac{|\mathcal{C}(A_t)|}{|\mathcal{C}|} \right), \quad (1)$$

where $\mathcal{C}$ are the 32 px cells of frame t and $\mathcal{C}(A_t)$ those containing a live track. The first term falls as K ’s content leaves the view (forward motion), the second as new content enters it (backward motion), so the measure is symmetric in effect. The next keyframe is the sharpest frame (Laplacian variance relative to a 2 s rolling median) among those with $\tau \leq \text{ov}(K, t) \leq \tau + \delta$ before the curve first drops below τ ; defaults $\tau = 0.75$, $\delta = 0.1$. If the overlap drops below τ within one step, the next frame is taken to keep the chain connected.

Multiplicity. Under uniform motion a point stays in view for about $1/(1 - \tau)$ keyframe spacings, so each surface point is observed by $m \approx 1/(1 - \tau)$ keyframes ($m = 4$ at $\tau = 0.75$) — the “two or more frames with a proper threshold of similarity” that triangulation needs, stated as a design parameter. We report the measured multiplicity, $1 + 2\bar{L}$ with $\bar{L}$ the mean forward track length in keyframes.

Cost. Only consecutive pairs are flowed, so analysis is linear in video length. An overlap matrix $M_{i,d} = \text{ov}(i, i+d)$ for all frames and $d \leq d_{\max}$ is built in one sweep: a fresh grid starts at every frame and all live grids advance together in a ring of $d_{\max}$ slots, one batched `grid_sample` per frame and a single read-back; selection is then a CPU walk over M .

3.4 Is it 3D? A parallax test

For each keyframe i we take its widest partner $j \leq i+6$ with $\text{ov} \geq 0.3$, compute RAFT flow *directly* between I_{K_i} and I_{K_j} in both directions, and keep consistent grid correspondences $(\mathbf{x}_n, \mathbf{y}_n)$. On a batch of all pairs we fit, by normalised DLT [15] with Cauchy IRLS, a homography H and a rank-2 fundamental matrix F . The residual $\|\mathbf{y} - H\mathbf{x}\|$ is rotation-compensated parallax: zero for pure rotation or a planar scene, growing with depth variation times baseline. We estimate the per-coordinate noise $\hat{\sigma} = 1.4826$ median $\sqrt{e_F^2}$ from F ’s Sampson residuals e_F — pure noise whenever F is correct, which includes degenerate motion — and normalise

$$\text{SNR} = \frac{\text{median}_n \|\mathbf{y}_n - H\mathbf{x}_n\|}{\sqrt{2}\sqrt{2\ln 2} \hat{\sigma}}, \quad (2)$$

where the denominator is the median transfer residual that isotropic noise in both views leaves, so $\text{SNR} = 1$ means no measurable parallax. A pass is *3D* when its median SNR over keyframe pairs is at least 2 and GRIC [6] (both models scored on geometric error with $\hat{\sigma}$) prefers F on at least 60% of pairs; *degenerate* when the SNR is below 1.5 and its 90th percentile below 2.

Why direct flow. Chaining consecutive flows makes long-range correspondence cheap but its error accumulates with a systematic per-step bias (§4, Fig. 2); over a keyframe span the drift reaches the size of the parallax being measured and is correlated across points, which the extra freedom of F partly absorbs. Direct flow keeps the error near 0.1 px at the same gaps.

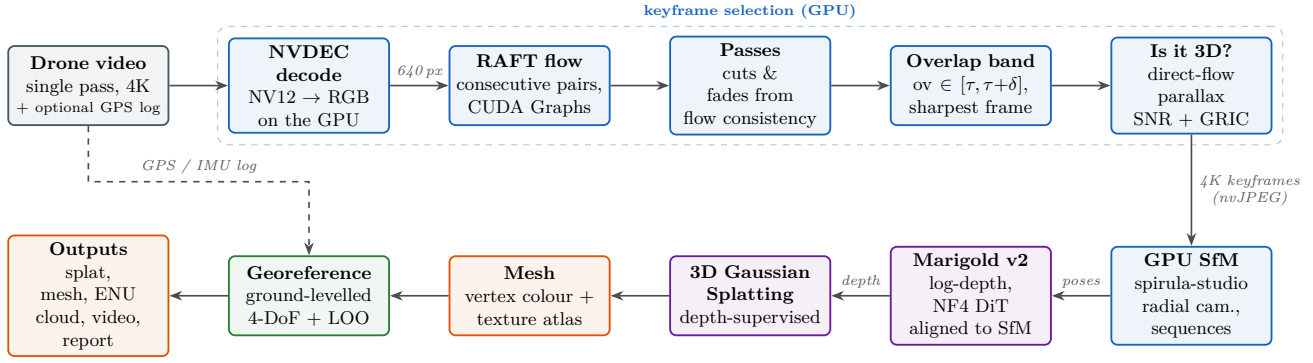


Figure 1: Pipeline. Everything between the video and the outputs runs on one GPU; stages exchange files only, so any stage can be re-run alone.

3.5 Structure from motion

Keyframes are written per pass (`images/pass_k/`) and reconstructed by spirula-studio’s GPU SfM with each pass declared a sequence, so neighbours are matched and trusted first, plus GPU pair selection for loop closure. We use a *single-focal* radial camera per pass: drone cameras have square pixels, and with independent f_x, f_y the weak geometry of forward or orbital motion let the two drift 10–20% apart for no gain in reprojection error (§4). Separate passes of an edited video become separate models unless they share views.

3.6 Monocular depth prior

Marigold v2 [5] predicts, per keyframe, $p(\mathbf{u})$ with $\log d(\mathbf{u}) = ap(\mathbf{u}) + b$ for unknown a, b . It runs as upstream: 4-bit NF4 base transformer (what its LoRA was trained on), rank-128 LoRA and fine-tuned VAE decoder, one rectified-flow step, bf16 autocast, at 1024 px; the quantised base is cached so a load reads ≈ 11 GB rather than re-quantising 39 GB. For every registered image the SfM tie points give depths z_n at pixels $\mathbf{u}_n$.

Affine is not enough at landscape scale. A Cauchy-IRLS fit of $\log z_n = ap(\mathbf{u}_n) + b$ works well where the tie points span a narrow depth range, but on drone footage the prior *compresses the far field*: on Jal Mahal it places the far shore at the depth of the palace and the mountain ridge barely beyond it (Fig. 3), so a single slope cannot fit both. We therefore calibrate each image with a monotone map: medians of $\log z$ in quantile bins of p , made non-decreasing by weighted pool-adjacent-violators, linear with the affine slope outside the knots. It keeps the prediction’s ordering and lets the mapping bend. Both calibrations are scored on *held-out* tie points (5-fold cross-validation per image).

Where the prior saturates. In a distant pass every tie point may sit where the prediction has saturated at its far end; the correlation between p and $\log z$ is then near zero or negative. The robust fit returns a non-positive slope and the stage writes no depth map for that image, so training is not supervised with wrong geometry.

The calibrated linear depth supervises splat training through spirula’s Pearson-correlation depth loss (weight 0.05). Because that loss is affine-invariant in *linear* depth, the calibration — which only SfM can provide — shapes the supervision itself, not merely its reporting. *What* is supervised matters as much: outside the range of predictions the tie points cover (sky, featureless far field) the calibration only extrapolates, and a depth invented for the sky asks the trainer to put splats there. The mask therefore keeps pixels whose prediction lies within the 0.5–99.5% range of the tie-point predictions (plus 5% margin) and below $3\times$ the 99th-percentile SfM depth. The canonical run below was trained before this mask existed (only the far cut, which removed nothing); the masked maps are evaluated as an ablation (§4).

3.7 Gaussian splatting and meshing

Each SfM model with at least 8 images trains a 3DGS model in spirula-studio (30k steps, half resolution, bilateral-grid exposure correction), with every 8th keyframe held out for evaluation; meshes are extracted from the trained splats with vertex colour and texture atlases.

3.8 Georeferencing a straight pass

With GPS positions $\mathbf{g}_n$ (local ENU) for camera centres $\mathbf{c}_n$, the usual fit is the 7-DoF similarity of Umeyama [16]. If the $\mathbf{c}_n$ are nearly collinear the rotation about the line is unobservable and GPS noise sets it; the in-sample RMSE still looks like GPS noise because the unobservable direction does not affect it. We instead estimate the model’s ground normal $\mathbf{n}$ by RANSAC on the sparse points (oriented towards the cameras), rotate $\mathbf{n}$ to $+z$, and fit yaw, scale and translation — scale and yaw by complex least squares on horizontal coordinates, the vertical offset as a mean residual. The full similarity is used only if the track spans two dimensions *and* its tilt against $\mathbf{n}$ exceeds 5° by three jackknife standard errors, i.e. GPS proves the ground is sloped. We report leave-one-out RMSE rather than in-sample RMSE, and a jackknife relative standard error of the scale as the scale check.

3.9 Keeping the GPU busy

Stock torchvision RAFT rebuilds its correlation lookup offsets on the CPU in each of its 12 refinement iterations, divides by a CPU tensor, builds coordinate grids on the CPU, and upsamples the flow after every iteration although inference keeps only the last; at 640px a call cost roughly the same whatever the batch size, a sign of launch- and sync-bound execution. Removing these (identical arithmetic, differences $\leq$ bf16 noise) and replaying the network from a CUDA Graph raised throughput —fold. The selector’s tracking avoids boolean-mask indexing, whose `nonzero` forces a host sync, in favour of scatter and redirected writes. Every stage runs under an NVML sampler recording utilisation, power, memory and the presence of other processes on the GPU, so no timing is reported without its context.

4 Experiments

Setup. One NVIDIA A100 80 GB PCIe, shared with other users’ jobs throughout (each stage’s record states the GPU’s occupancy); PyTorch 2.14 with CUDA 13.2, Python 3.14, ffmpeg 9.0 (NVDEC/NVENC), spirula-studio 2026.9.24 (Vulkan backend). Data: fifteen public drone videos (Table 1); they carry no flight logs.

Table 1: Sample footage (public YouTube re-encodes, VP9).

Video	Resolution	fps	s
ANGKOR WAT Breathtaking Cinemati	3840×2160	24.00	193
Above clouds	2560×1440	50.00	115
Aerial Views of Rural Riches	2560×1440	23.98	19
Breathtaking Aerial Views of Hanoi	3840×2160	29.97	103
Colosseum like never before	3840×2160	29.97	135
Eiffel Tower Drone 4k	3840×2160	29.97	306
FPV Drone Flight through Beautiful	3840×2160	30.00	67
GoPro	1080×1920	29.97	18
HAGIA SOPHIA 4K	3840×2160	29.97	85
Jal Mahal, Jaipur, India, Drone Ci	3840×2160	24.00	55
Notre Dame Drone Paris 4k	3840×2160	29.97	214
Petronas Twin Tower in FPV drone	3840×2160	59.94	35
Qutub Minar, New Delhi, India 4K D	3840×2160	29.97	187
Reichstag Berlin in 4K	3840×2160	30.00	130
The Messiah	3840×2160	29.97	94

4.1 Passes, keyframes and the 3D verdict

On Jal Mahal the flow signal splits the edit into 5 passes at its cuts and fades, and the band selects 213 keyframes; the mean overlap with the previous keyframe lies in 0.74–0.86, inside or just above the target band, and every pass is judged *3d* (Table 2). Decoding and flow for the 658 analysis frames took 37 s, selection 12 s and 4K extraction 45 s.

Negative control. A synthetic clip of a camera rotating about its centre over one real 4K frame (`tools/make_degenerate_clip.py`) has no parallax by construction. It yields SNR 1.24 (median residual 0.02°) and the verdict *degenerate*; GRIC nonetheless preferred

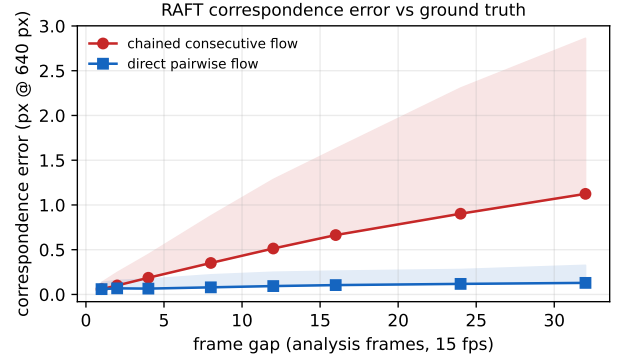


Figure 2: Correspondence error against exact ground truth on the rotation control. Chained consecutive flow drifts almost linearly (1.12 px median at 32 frames); direct pairwise flow stays at 0.13 px.

F on 60% of its pairs, which is why GRIC only confirms the SNR decision.

4.2 Structure from motion

With the same SfM settings, 1 fps sampling registers 40 of 55 frames in 2 disconnected models (the largest 22), and uniform sampling at the selector’s budget of 241 frames registers 237 but splits into 4 models (largest 98). The overlap band registers all 241 (largest 155); after cropping the letterbox and trimming fades the canonical run registers 213 of 213 (Table 4). A model per group of passes is expected: the passes of a cinematic edit are cut from different flights and need not overlap. One focal length per pass keeps the reprojection error while OpenCV’s independent f_x, f_y drift apart (Table 5).

4.3 Monocular depth prior

Table 6 scores each calibration on tie points it was not fitted to. The monotone map lowers held-out AbsRel on every model with depth; the gain is largest where the scene spans a wide depth range (Fig. 3), which a single log-affine map cannot follow. Model 3 (30 keyframes, the far lake-side pass) has no image whose prediction correlates positively with its tie-point depths; it is trained without depth rather than with a wrong one.

4.4 Gaussian splatting

4.5 Georeferencing a straight pass

None of the sample videos carries a flight log, so georeferencing is evaluated in simulation (`experiments/georef_study.py`): a 300 m track at 80 m altitude over flat ground with buildings, the SfM model hidden behind a random similarity, horizontal GPS noise, and a sideways deviation of the track from a straight line of 0–60 m; the error is measured at ground points 100 m off the track. (Sloped ground, where the rule falls back to the similarity, is covered by the unit tests.) The levelled fit is never

Table 2: Pass segmentation and keyframes per run. Overlap: mean co-visibility with the previous keyframe; views: measured views per point; SNR: median direct-flow parallax SNR (1 = none).

Run	Pass	Time (s)	KF	Overlap	Views	SNR	Verdict
jal_mahal	0	0.0–7.4	30	0.77	6.9	22.4	3d
jal_mahal	1	8.0–14.1	14	0.79	5.2	3.6	3d
jal_mahal	2	14.2–23.8	59	0.77	7.7	6.0	3d
jal_mahal	3	23.8–36.0	97	0.74	7.4	16.3	3d
jal_mahal	4	37.2–54.0	13	0.86	8.2	3.2	3d

Table 3: Structure from motion on the selected keyframes.

Run	KF	Registered	Models	Reproj. (px)	Time (s)
jal_mahal	213	213	4	0.79	255

Table 4: Frame sampling on Jal Mahal (same SfM settings).

Sampling	Frames	Registered	Models	Largest
1 fps (uniform)	55	40	2	22
uniform, same budget	241	237	4	98
overlap band (dev)	241	241	4	155
overlap band (canonical)	213	213	4	156

worse than the similarity (worst 2.1 m) and its metric scale error stays below 0.6 % (Table 8, Fig. 4).

4.6 Throughput and GPU use

Table 6: Marigold v2 against SfM tie points, scored on held-out tie points (5-fold cross-validation per image, median over images): log-affine vs monotone calibration.

Run	Model	Aligned	AbsRel aff.	AbsRel mono.	δ_1 aff.	δ_1 mono.
jal_mahal	0	156/156	0.050	0.027	0.857	0.886
jal_mahal	1	14/14	0.009	0.006	1.000	1.000
jal_mahal	2	13/13	0.075	0.033	0.945	0.989
jal_mahal	3	0/30	—	—	—	—

Table 5: Camera model on Jal Mahal: independent f_x, f_y (OpenCV) against one focal (radial).

Camera	Focal(s) per pass (px)	Reproj. (px)	Points
OpenCV (fx, fy free)	3112/2812, 2055/2080, 2827/3382, 2986/2754	0.761	96,940
radial (one focal)	3118, 2684, 2808, 3109	0.771	96,598

Table 7: 3DGS on held-out keyframes (every 8th keyframe never used in training).

Run	Model	Splats	PSNR	SSIM	LPIPS	Train (s)
jal_mahal	model_0, depth prior (v1 mask)	3,000,000	35.00	0.977	—	3822

Table 8: Georeferencing study (medians over 20 trials): off-track error of the similarity and of our rule, in-sample RMSE of the similarity, held-out RMSE of ours, metric scale error.

GPS σ (m)	Lateral (m)	Sim. (m)	Ours (m)	Sim. in-sample	Ours LOO	Scale err.
0.5	0	149.9	0.21	0.71	0.73	0.1%
0.5	5	5.5	0.19	0.69	0.71	0.0%
0.5	20	2.1	0.19	0.68	0.70	0.0%
0.5	60	0.7	0.17	0.72	0.74	0.1%
1.5	0	180.0	0.62	2.08	2.16	0.1%
1.5	5	19.5	0.54	2.10	2.18	0.1%
1.5	20	5.0	0.58	2.16	2.24	0.2%
1.5	60	1.9	0.63	2.08	2.16	0.2%
3.0	0	220.7	0.92	4.15	4.29	0.2%
3.0	5	27.0	1.11	4.00	4.15	0.2%
3.0	20	7.6	1.29	4.28	4.44	0.3%
3.0	60	2.8	1.14	4.29	4.44	0.2%
5.0	0	146.9	1.77	7.08	7.32	0.3%
5.0	5	24.4	2.13	7.13	7.37	0.5%
5.0	20	11.0	1.59	6.92	7.14	0.4%
5.0	60	5.4	2.08	6.83	7.07	0.6%

Table 9: Per-stage wall time and GPU use (NVML, 0.5 s samples). Own/host: peak GPU memory and host RSS of our process tree.

Run	Stage	s	Util.%	W	Own GPU (GB)	Host (GB)	Shared
jal_mahal	keyframes	109	83	256	10.6	6.3	yes
jal_mahal	sfm	256	83	213	9.9	4.2	yes
jal_mahal	depth	522	84	265	24.3	11.2	yes

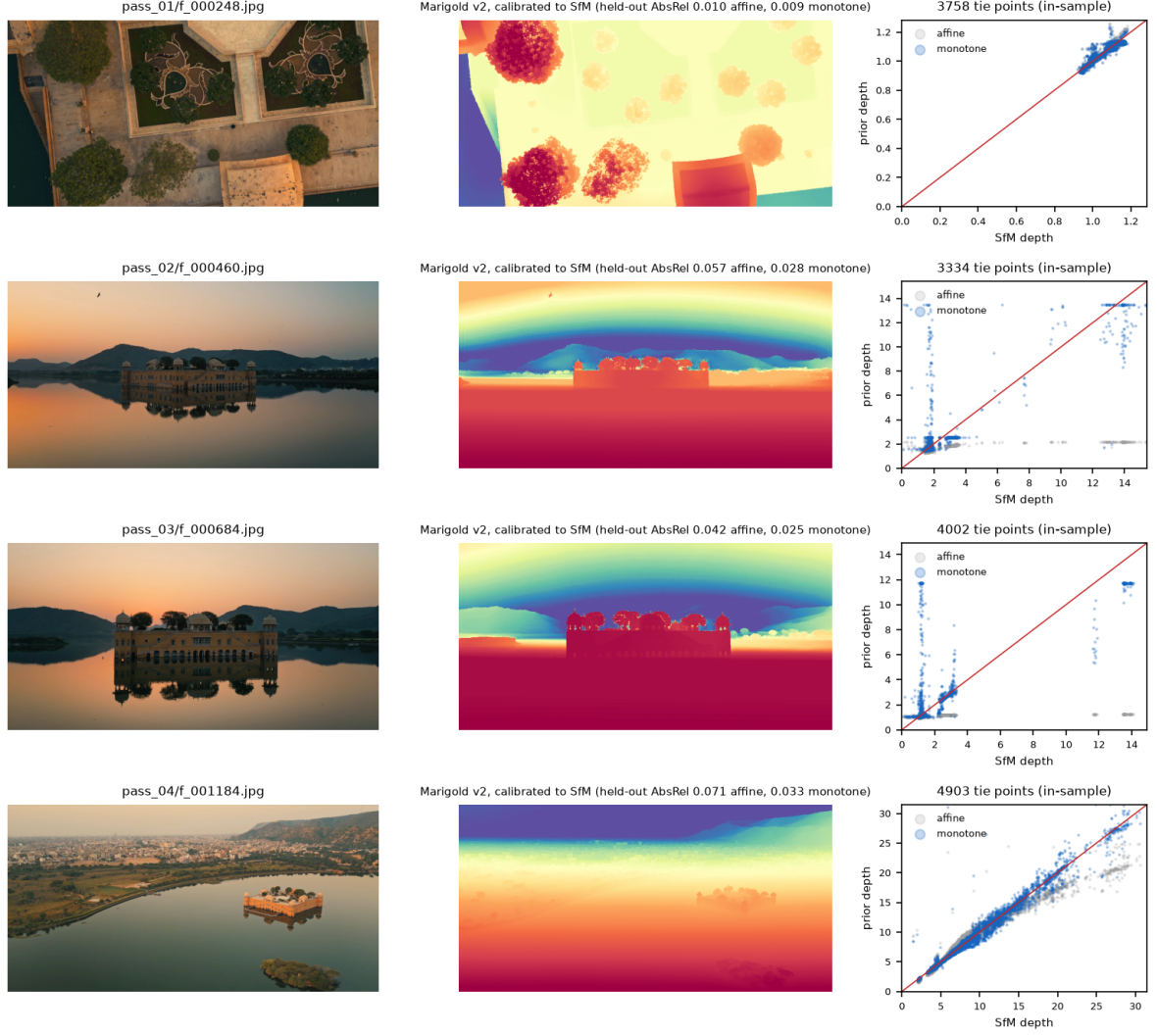


Figure 3: Marigold v2 on Jal Mahal keyframes (canonical run). Right: prior depth against SfM tie-point depth for the log-affine fit (grey) and the monotone calibration (blue). On the lake-facing passes the affine fit flattens the far shore and the ridge; the monotone map follows them (it also sends a few near points far). Held-out AbsRel for both is given above each depth map.

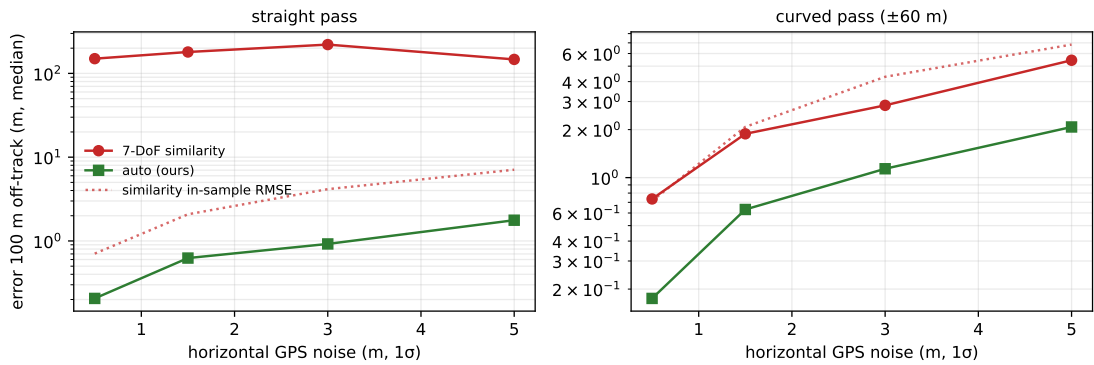


Figure 4: Error of ground 100 m off-track after GPS alignment (median of 20 synthetic scenes per point). The 7-DoF similarity is off by up to 221 m on a straight pass while its in-sample RMSE (dotted) only reflects GPS noise; the ground-levelled fit stays near the GPS noise.

5 Limitations

No flight logs in the public samples. The fifteen sample videos are YouTube re-encodes without GPS; georeferencing is therefore evaluated in simulation (§4), and the georef stage runs on real data only when a DJI SRT, CSV, GPX or JSON log is supplied. **Edited footage.** The samples are edits of several shots; each pass becomes its own model, which a true single pass would not need. **Water and sky.** Reflections are reconstructed as mirrored geometry below the surface, as 3DGS always does, and single-frame flow-consistency dips on water add keyframes. **Shared GPU.** Timings were taken while other users’ jobs shared the A100; each stage’s record says so. **What the scores measure.** Held-out PSNR compares renders with keyframes between training views; it does not measure geometry away from the flight path, and without survey data we report no absolute accuracy for the meshes. **Dynamic objects.** Cars, boats and people are reconstructed as blur or floaters; masking them is open. **Compositional objects.** WorldSculpt [14] could turn the splat scene

into per-object meshes given instance masks and boxes; we have not integrated it.

6 Conclusion

A single drone pass can be turned into a splat model and mesh on one GPU, provided the system chooses its frames by what they see rather than by the clock, checks that the motion contains parallax before trusting it, and aligns to GPS with the constraint a straight line cannot provide. The code, configurations and the scripts that regenerate every number in this paper are in the repository.

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